

Problem Set 7Question 1

IS curve $y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \varepsilon_t^y$ (1)

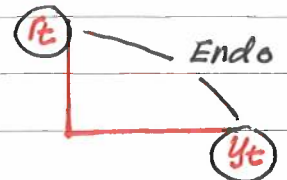
Note that since $i_t = r_t + \pi_t$, then IS curve can also be written as

$$y_t = y_t^* - \alpha(r_t - r^*) + \varepsilon_t^y \quad (1a)$$

where α is the sensitivity of y_t to change in r_t .

$$\Delta y_t = \Delta y_t^* - \alpha \Delta r_t + \alpha \Delta r^* + \Delta \varepsilon_t^y$$

Slope of IS = $\frac{\Delta r_t}{\Delta y_t} = -\frac{1}{\alpha}$



Endo variables : r_t and y_t (IS is on $y_t - r_t$ space)

Exo variables : y_t^* , r^* and ε_t^y

For example, when $\varepsilon_t^y \uparrow \rightarrow \Delta \varepsilon_t^y > 0$. Then $\Delta y_t = \Delta \varepsilon_t^y > 0 \therefore y_t \uparrow$

Given r_t , $y_t \uparrow \rightarrow \overrightarrow{IS}$.

↙ Inflation target

MPR $i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*)$ (2)

and β_π shows how its response to π_t i.e. then β_π measures the sensitivity of i_t to change in π_t / $(\pi_t - \pi^*)$

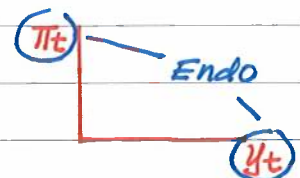
(a) IS-MPR is derived by substituting (2) into (1)

$$\begin{aligned} y_t &= y_t^* - \alpha[r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) - \pi_t - r^*] + \varepsilon_t^y \\ &= y_t^* - \alpha(\pi^* + \beta_\pi \pi_t - \beta_\pi \pi^* - \pi_t) + \varepsilon_t^y \\ &= y_t^* - \alpha(1 - \beta_\pi)\pi^* - \alpha(\beta_\pi - 1)\pi_t + \varepsilon_t^y \\ y_t &= y_t^* - \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \varepsilon_t^y \quad (3) \end{aligned}$$

And when we totally differentiate (3) we get

$$\Delta y_t = \Delta y_t^* - \alpha(\beta_\pi - 1)(\Delta \pi_t - \Delta \pi^*) + \Delta \varepsilon_t^y$$

Slope of IS-MPR = $\frac{\Delta \pi_t}{\Delta y_t} = -\frac{1}{\alpha(\beta_\pi - 1)} < 0$ if $\beta_\pi > 1$



Endo variables : π_t and y_t

Exo variables : y_t^* , π^* and ε_t^y

$\beta_\pi > 1$ b/c $i_t = r_t + \pi_t \therefore \Delta i_t = \Delta r_t + \Delta \pi_t$ and $\Delta r_t = \Delta i_t - \Delta \pi_t$

Suppose when $\pi_t \uparrow$ but $\overline{i_t}$, then $r_t \downarrow \therefore \Delta r_t = -\Delta \pi_t < 0$

Note that from MPR $\Delta i_t = \Delta r^* + \Delta \pi^* + \beta_\pi \cdot \Delta \pi_t - \beta_\pi \cdot \Delta \pi^*$

If r^*, π^* are constant, then

$$\Delta i_t = \beta_\pi \cdot \Delta \pi_t$$

Sub this into Δr_t $\Delta r_t = \Delta i_t - \Delta \pi_t$

$$\Delta r_t = \beta_\pi \cdot \Delta \pi_t - \Delta \pi_t \therefore \Delta r_t = (\beta_\pi - 1) \cdot \Delta \pi_t$$

$$\Delta r_t = (\beta\pi - 1) \cdot \Delta\pi_t$$

When $\pi_t \uparrow$, $r_t \uparrow$ only if $\beta\pi - 1 > 0 \therefore \beta\pi > 1$.

This means that when $\pi_t \uparrow$, $r_t \downarrow$ if $\beta\pi < 1$. In order to ensure that when $\pi_t \uparrow$, $r_t \uparrow$, it must increase by more i.e. $\beta\pi > 1$.

For IS-MPR, when $\pi_t \uparrow \xrightarrow{\beta\pi > 1} i_t \uparrow \xrightarrow{\beta\pi > 1} r_t \uparrow \rightarrow COB \uparrow \begin{cases} C_t \downarrow \\ I_t \downarrow \end{cases} \rightarrow y_t \downarrow$
IS-MPR is downward.

$\pi_t \downarrow$ and $\rightarrow \dots$ the system is stable.

However, when $\pi_t \uparrow \xrightarrow{\beta\pi < 1} i_t \uparrow \xrightarrow{\beta\pi < 1} r_t \downarrow \rightarrow COB \downarrow \begin{cases} C_t \uparrow \\ I_t \uparrow \end{cases} \rightarrow y_t \uparrow \rightarrow \pi_t \uparrow$

the system is explosive (not stable)

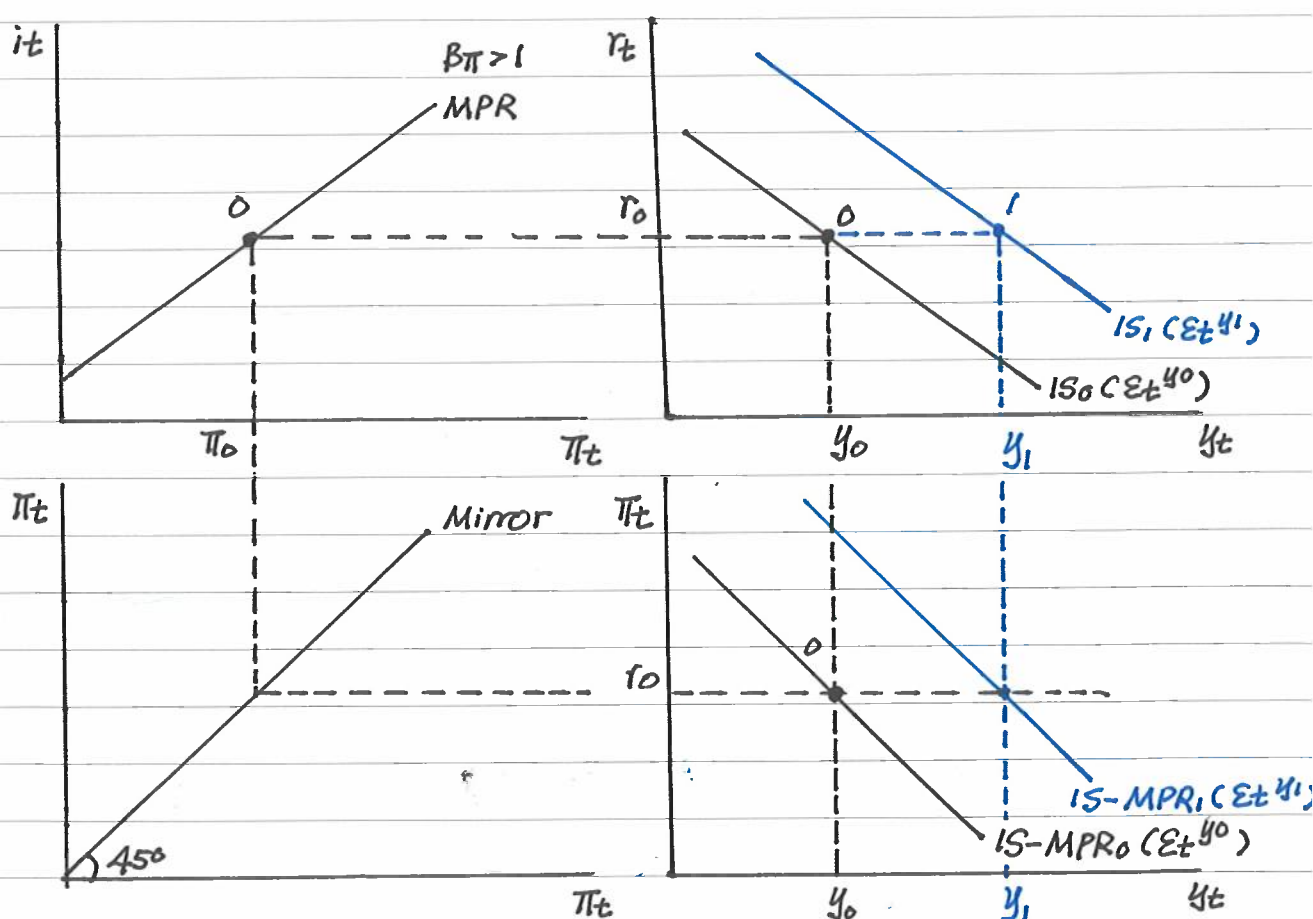
(b) Since $\Delta y_t = \Delta y_t^* - \alpha(\beta\pi - 1)\Delta\pi_t + \alpha(\beta\pi - 1)\Delta\pi^* + \Delta\epsilon_t^y$

Exo variables: y_t^* , π^* and ϵ_t^y .

Change in exo variable(s) will shift the IS-MPR.

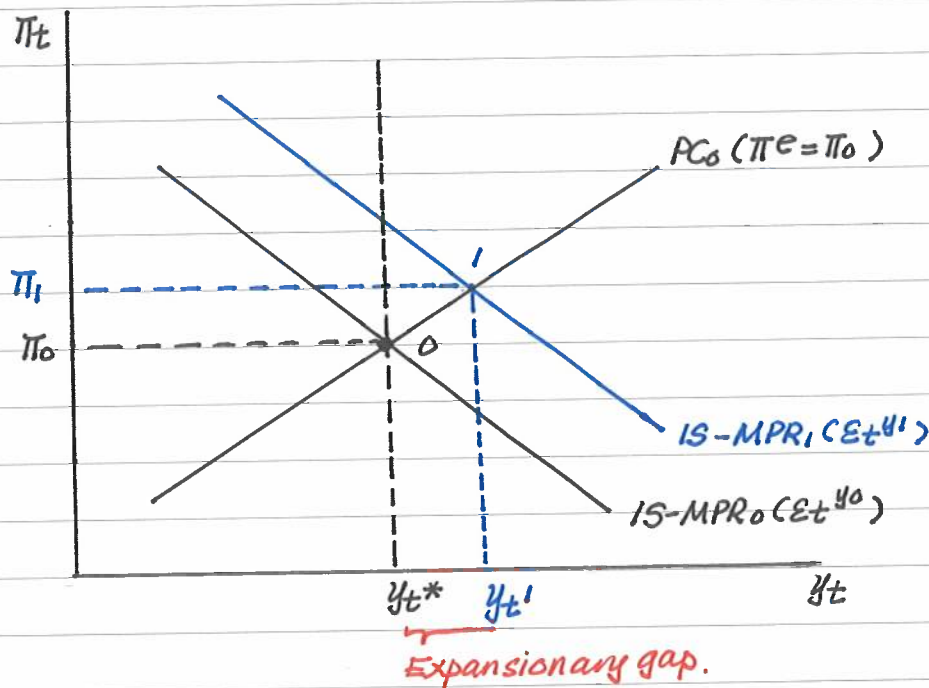
Example: $\epsilon_t^y \uparrow \rightarrow \Delta\epsilon_t^y > 0$. All else constant.

Then, $\Delta y_t = \Delta\epsilon_t^y > 0 \rightarrow y_t \uparrow$ given $\pi_t \rightarrow$ IS-MPR



Question 2

(a) Consumer confidence $\uparrow \rightarrow C_t \uparrow \rightarrow \varepsilon_t^y \uparrow \rightarrow \Delta \varepsilon_t^y > 0 \rightarrow \Delta y_t = \Delta \varepsilon_t^y > 0 \rightarrow y_t \uparrow$ and $\overrightarrow{IS}$ and $\overrightarrow{IS-MPR}$



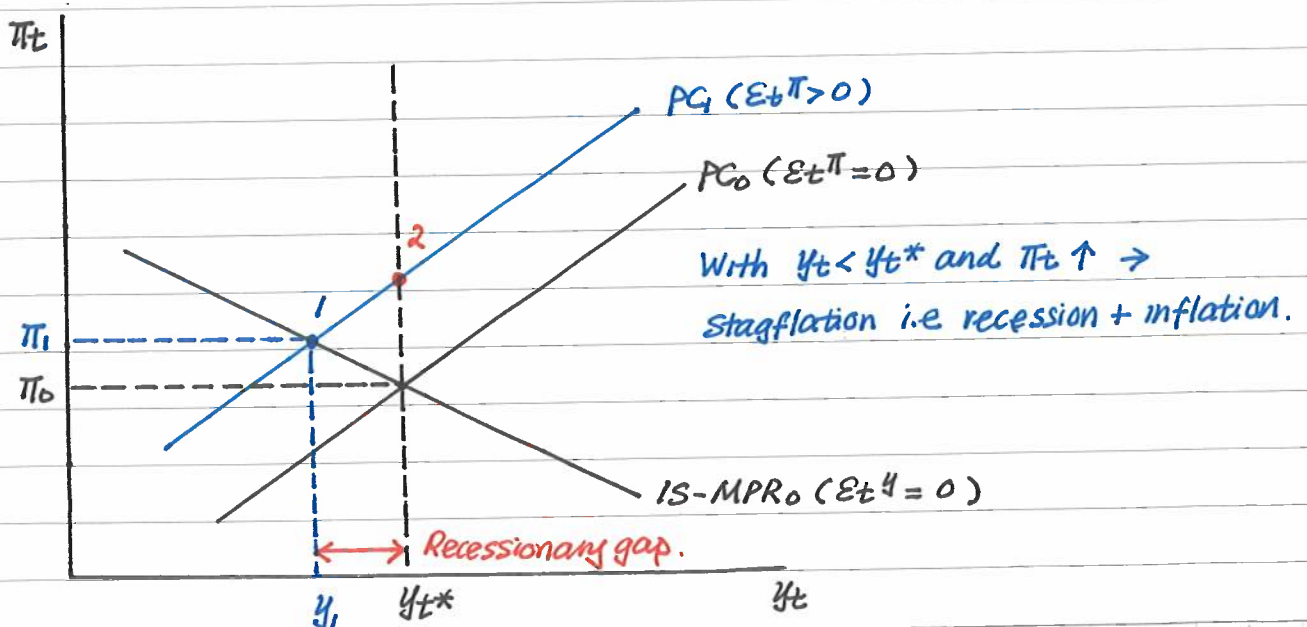
(b) Since $y_t^1 > y_t^* \rightarrow$ expansionary gap.

$C_t \uparrow \rightarrow \varepsilon_t^y \uparrow \rightarrow y_t \uparrow \rightarrow y_t > y_t^*, (y_t - y_t^*) > 0 \rightarrow \pi_t \uparrow$.

Note that PC is $\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t \pi$

Question 3

$\varepsilon_t \pi \uparrow \rightarrow \Delta \pi_t = \Delta \varepsilon_t \pi > 0, \pi_t \uparrow$ given $y_t \rightarrow$ PC shifts up.



Question 4

(a) MPR $i_t = r^* + \pi^* + \beta\pi(\pi_t - \pi^*)$

Note that $\Delta i_t = \Delta\pi^* - \beta\pi \cdot \Delta\pi^*$

$\Delta i_t = (1 - \beta\pi) \cdot \Delta\pi^*$

$\Delta i_t = -(\underbrace{\beta\pi}_{+} - \underbrace{1}_{+}) \cdot \Delta\pi^* < 0$

$\therefore$ Increasing π^* i.e. $\uparrow$ inflation target.

When $i_t \downarrow \rightarrow r_t \downarrow \rightarrow COB \downarrow \rightarrow \begin{cases} C_t \uparrow \\ I_t \uparrow \end{cases} \rightarrow y_t \uparrow \rightarrow$ IS-MPR Diagram 2(a)

